

TOPICAL REVIEW

A comprehensive review on material selection, processing, characterization and applications of aluminium metal matrix composites

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A comprehensive review on material selection, processing, characterization and applications of aluminium metal matrix composites

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V Mahesh Kumar  and C V Venkatesh

Department of Mechanical Engineering, Malnad College of Engineering, Hassan, Karnataka, India

E-mail: mareshkumarv89@gmail.com

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Abstract

Aluminium alloy based metal matrix composites have found increased applications in space, automobile, aircraft structures because of simplicity of operation and reasonably good mechanical properties. But the main drawbacks in aluminium matrix metal matrix composites (AMMCs) are distribution of particles, wettability and chemical reaction between reinforcement and matrix. These factors have considerable influence on mechanical, tribological and corrosion properties of the developed metal matrix composites. Therefore it is necessary to overcome these drawbacks by optimizing the process parameters to produce AMMCs tailored to an application. This review focuses on to the issues related to the processing of MMCs, selection of both matrix and reinforcement, techniques available for the characterization of AMMCs. With the available literature an attempt is made to provide the guidelines for researcher about present scenario of AMMCs fabrication, properties and applications.

1. Introduction

Metal matrix composites consist of a metal of low density, such as Aluminium or Magnesium, reinforced with particulate or fibres of a Ceramic material, such as Silicon Carbide or Graphite. Compared to pure metals, MMCs offer higher specific strength and stiffness; higher operating temperature and greater resistance to wear as well as these properties can be tailored to suit a particular application. An enormous amount of work has been carried out in an effort to characterize the mechanical behaviour of particle reinforced MMCs. These shows a significantly higher stiffness and mechanical strength compared to matrix alloys, but often suffers from lower ductility and inferior fracture toughness. Also shows improved properties under relatively higher tensile and compressive forces by the transfer and distribution of an applied load from the ductile matrix to the reinforcement material.

There are different processes available to manufacture MMCs, and among all the liquid-state processes, stir casting is considered as the most potential method of manufacturing for engineering applications in terms of production capacity and cost considerations. This casting technique is economical, easier to apply and more convenient for mass production compared to other manufacturing techniques.

The pure aluminium alloy properties can be enhanced by incorporating the most commonly used reinforcement types such as fibres, particulates etc. The influence of copper coated steel fibres shows enhanced wear properties and significant improvement in hardness, addition of 5% copper coated steel fiber shows hardness value doubled compared with unreinforced pure aluminium [1]. The addition of boron carbide in Al 2024 matrix produced by stir casting process shows enhanced physical and mechanical properties than unreinforced alloys but the density of the composites shows decreasing trend with increasing volume fraction and decreasing particle size [2]. The Boron carbide and molybdenum disulphide particles are added to Al 2219 processed using stir casting technique shows a random and fine dispersement in matrix alloy, density and microhardness of the composite increases as compared to Al 2219 alloy [3]. Al₂O₃ reinforced in AA2024 shows a

Table 1. Commercially available aluminium 1xxx series alloys [14].

Registered international designation	Composition								
	Cu	Fe	Si	Zn	Mg	Mn	Ti	V	Others
1030	0.10	0.6	0.35	0.10	0.05	0.05	0.03	0.05	0.03
1035	0.10	0.6	0.35	0.10	0.05	0.05	0.03	0.05	0.03
1040	0.10	0.50	0.30	0.10	0.05	0.05	0.03	0.05	0.03
1045	0.10	0.45	0.30	0.05	0.05	0.05	0.03	0.05	0.03
1055	0.05	0.40	0.25	0.05	0.05	0.05	0.03	0.05	0.03
1075	0.04	0.20	0.20	0.04	0.03	0.03	0.03	0.05	0.03
1095	0.010	0.040	0.030	0.010	0.010	0.010	0.005	—	0.005

superior fatigue behaviour compared to unreinforced Al 2024 [4]. The silicon particles incorporated in Al–Cu–Mg alloy using powder metallurgy process shows reduced wear loss with increasing applied load and sliding speed [5]. Graphite and silicon carbide addition with Al–Si alloys shows better wear characteristics compared to the composites that with reinforcements to the same matrix individually [6]. Magnesium based metal matrix composites reinforced with silicon carbide shows superior wear properties under the lower load and the results shows that it is not suitable for higher load and higher speed applications [7]. The addition of beryl particles on Al6061 through a liquid metallurgy process, the results revealed that addition of beryl particles of about 10% can bring significant changes in the composites, it is found to have 15.38% improvement in tensile strength and 8.9% decrement in specific wear rate [8]. The use of Red mud which is the industrial waste as the reinforcement shows improved wear properties over the aluminium alloy matrix and it can replace the use of expensive reinforcement materials such as SiC and Al_2O_3 [9]. The corrosion resistance of Al–Si–Mg alloys can be significantly improved when it is reinforced with both zircon and alumina particles [10]. The influence of *in situ* formed particulate reinforced aluminium alloys shows significant refinement in the microstructure and enhanced mechanical properties [11]. As high specific strength and damage tolerance becomes great importance in automotive and aerospace applications, the use of nano particulate fillers becomes more popular because it shows high strength, young's modulus, good ductility and lower density [12]. Stir casting is found to be the best process to produce the metal matrix composites, the optical and scanning electron micrographs shows a homogeneous dispersion of the added reinforcements [13].

2. Review on aluminium based composites

2.1. Aluminium 1xxxseries

The 1xxx series alloys comprise of Aluminium 99% or higher purity and this series has low mechanical properties and extremely high corrosion resistance, workability as well as high thermal and electrical conductivity. The composition of commercially available aluminium 1xxx series is shown in the table 1.

The strength properties of pure aluminium composites can be improved by reinforcing it with hard ceramic particles, the addition of Titanium oxide (TiO_2), Titanium carbide (TiC) and Titanium nitride (TiN) results in an enhanced mechanical and surface properties compared to composites reinforced individually. SEM micrographs of Al–15TiC and Al–15TiN are shown in figures 2(a) and (b) [15]. The aluminium alloy is reinforced with Al_2O_3 produced using powder metallurgy process, identification of these particles and their composition done using EDS spectrum analysis is as shown in figures 1(a) and (b). The tribological property of the composites improves with increasing alumina and compaction pressure [16]. In case of Aluminium matrix composites with varying percentage of SiC particles prepared using the stir casting process, results revealed that with the incorporation of SiC particles increased tensile strength and hardness and recorded maximum value at 20 wt% addition [17]. The study on Al_2O_3 reinforced aluminium alloy shows a different trend in its physical and mechanical properties, the effect of sintering and mechanical mixing technique was used in producing the composites. The results shows that for the addition of 5% Al_2O_3 particles it recorded the highest hardness and compression strength beyond that both the value starts decreasing and also higher density and lower porosity was recorded, it is also observed that presence of pores has a negative effect on the hardness and thermal conductivity, as the porosity of the composites increases by increasing alumina wt% the hardness and thermal conductivity decreases [18]. The dry sliding wear behaviour of the graphite reinforced aluminium composites fabricated using molten metal mixing and tested by means of a pin-on-disc type wear testing machine, in addition to effect of volume fraction of reinforcement, contact stress, the effect of sliding distance and sliding velocities were also observed using analysis of variance approach. The results shows that the sliding distance has 74.65% of highest significance on wear loss, other factors are of sliding velocity of 11.37%, contact stress of 9.19% and it is recorded that reinforcement has least significance. The influence on co-efficient of friction is at

Table 2. Summary of materials, processing technique and its properties of Al lxxx series alloys.

References	Mohapatra <i>et al</i> [15]	Mandal A <i>et al</i> [16]	Rahman MH <i>et al</i> [17]	Mohammed MM <i>et al</i> [18]	Rajesh S <i>et al</i> [19]	Scudino S <i>et al</i> [20]
Matrix type	Pure aluminium alloy	Pure aluminium alloy	Pure aluminium alloy	Pure aluminium alloy	Pure aluminium alloy	Pure aluminium alloy
Reinforcement	TiO ₂ , micro Al ₂ O ₃ particles	Al ₃ Zr and Al ₂ O ₃ particles	SiC	Al ₂ O ₃	Graphite	Zr-based glassy particles
Reinforcement size	50 nm and 300 μ m	Al ₃ Zr–11–39 μ m Al ₂ O ₃ –30 μ m	53–74 μ m	5–10 μ m	50 μ m	—
Volume fraction	—	5, 10 and 20%	0, 5, 10 and 20%	0, 5, 10 and 15%	0 to 5% with increment of 1.25%	40%–60%
MMC Fabrication	powder metallurgy	powder metallurgy	Stir casting process	powder metallurgy	Stir casting process	powder metallurgy
Effect on its properties	Enhanced mechanical and surface properties.	hardness value is maximum even at 700°C sintering temperature, but wear loss is high at higher normal load and sliding distance.	Increased tensile strength and hardness	Enhances compressive strength upto 5% addition of Al ₂ O ₃ , but it shows negative effects on electrical and thermal conductivity	Sliding distance has highest significance on wear loss	Compressive strength increases by 28% of the base alloy for 40% addition of reinforcement

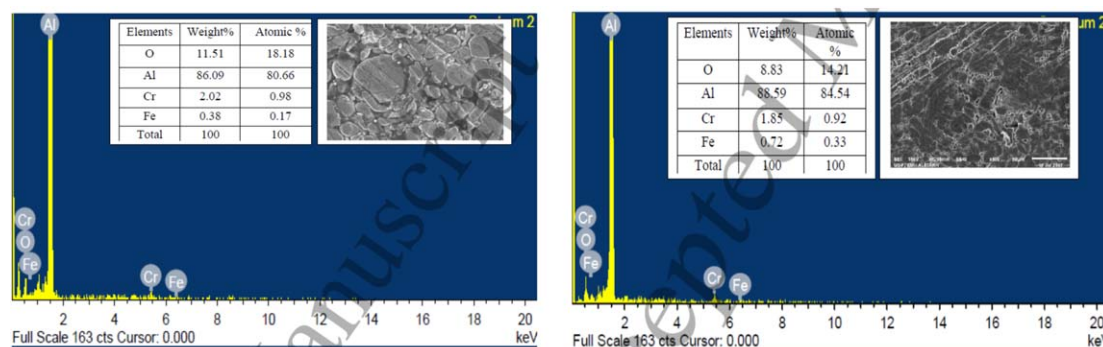


Figure 1. EDS of (a) green composite (Al-Fe-Cr-20 wt%Al₂O₃) (b) sintered composite (Al-Fe-Cr-20 wt%Al₂O₃) [16].

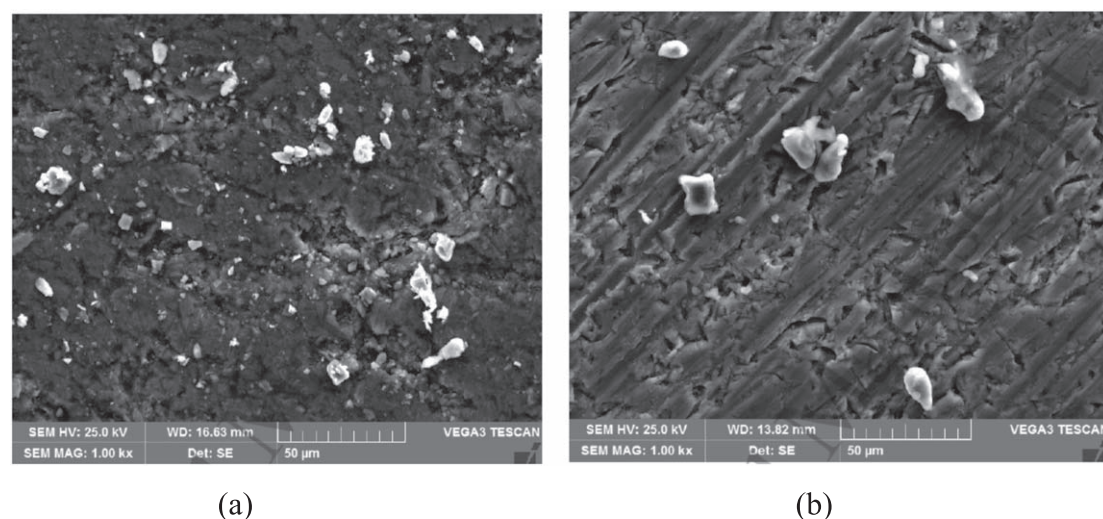


Figure 2. SEM images of (a) Al-15TiC (b) Al-15TiN [15].

reinforcement percentage is 45.59%, contact stress is 0.85% and sliding velocity is 0.46% [19]. The influence of Zr-based glassy particles on pure aluminium alloy processed using powder metallurgy, the main advantage of glassy powder is of the viscous flow behaviour in order to take this advantage these reinforcements are consolidated into highly dense bulk specimens by hot pressing followed by hot extrusion, compression tests conducted at room temperature results shows that the addition of the glass reinforcements is very effective for improving the mechanical properties compared to pure alloy. The compressive strength increases from 155 MPa for pure aluminium to 200 MPa for the composite with 40% weight fraction for the reinforcement [20]. The application of statistical approaches like grey-Taguchi method for optimization of abrasive wear properties of aluminium based composites and to study the effect of different parameters like contact stresses, volume fraction, sliding distance and sliding velocity. This is performed for SiC reinforced aluminium composites and the results indicated that that the sliding velocity was the most effective parameter among all followed by the volume fraction, sliding distance and contact stress. Delamination type of wear was observed using Scanning Electron Microscopy [21]. Table 2 summarises the materials, processing techniques and properties of Al 1xxx series alloys.

2.2. Aluminium 2xxx series

The 2xxx series are heat treatable, where copper is the principal alloying element often with magnesium as a secondary addition. The composition of commercially available aluminium 2xxx series is shown in the table 3.

The addition of SiC and flyash to Al 2024 matrix alloys through stir casting process resulting in an improvement of its tribological properties compared to unreinforced alloys. Scanning Electron Microscopy (SEM) studies shown in figures 3(a) and (b) confirms the uniform distribution of reinforcement particles [2]. It

Table 3. Commercially available aluminium 2xxx series alloys [14].

Registered international designation	Composition in Wt%								
	Cu	Si	Mg	Mn	Ni	Zi	Ti	Li	Zr
2008	0.9	0.65	0.38	—	—	—	—	—	—
2010	1.0	—	0.7	0.25	—	—	—	—	—
2014	4.4	0.8	0.50	0.8	—	—	—	—	—
2017	4.0	0.50	0.6	0.7	—	—	—	—	—
2020	4.5	0.40	0.03	0.6	—	0.25	0.10	1.4	—
2024	4.4	—	1.5	0.6	—	—	—	—	—
2025	4.4	0.8	—	0.8	—	—	—	—	—
2036	2.6	—	0.45	0.25	—	—	—	—	—
2080	3.6	0.10	1.8	0.25	—	0.10	—	—	0.15
2117	2.6	—	0.35	—	—	—	—	—	—
2124	4.4	—	1.5	0.6	—	—	—	—	—
2195	4.0	—	0.50	—	—	—	—	—	—
2219	6.3	—	—	0.30	—	—	0.06	—	—
2319	6.3	—	—	0.30	—	—	0.15	—	—
2618	2.3	0.18	1.6	—	1	—	0.07	—	—

was shown that specific wear rate of Al 2024 matrix can be improved by reinforcing Si_3N_4 reinforced processed by stir casting method, microscopic observation revealed that dispersion of the reinforcement particles was generally uniform with increasing weight fraction resulting in agglomeration of the particles and it shows that density and porosity of the composite increases by increasing reinforcing content as shown in figure 4 [22]. The statistical approach was conducted on SiC and Gr reinforced Al 2219 hybrid metal matrix composites by considering sliding velocity, load, sliding distance as the parameters, it was observed that Al–SiC–Gr composites exhibit higher wear resistance compared with Al–SiC reinforcement composites. Wear loss decreases with an increase in sliding velocity for the combination of composites, but increases with increase in load and sliding distance [23]. The comparative study was made to understand the better reinforcement among SiC whiskers, Al_2O_3 fibers and SiC particles, results revealed that particulates are the most beneficial for enhancing the wear resistance of composites. The whisker reinforcements are effective in reducing the steady state wear rates only when it is added in high volume fraction and it is very less effective for low volume fraction, fiber fragments become greater in size, so their aggressive action on the counter face material becomes heavier with increasing volume fraction of reinforcement, but the particle reinforcement, whose diameter is bigger than fibers and whiskers avoids initial severe wear and decreases the steady state wear rate considerably because of the large load supporting effect [24]. The Al–Cu–Mg based alloys were reinforced with 5 wt% SiC and Al_2O_3 particles processed through stir casting technique, it is recorded that a considerable improvement i.e. 16.4% in ultimate tensile strength of the matrix with the addition of these reinforcement and effect on ductility shows 31% higher in Al_2O_3 reinforced composites compared to SiC reinforced composites than the matrix alloy [28]. The ceramic reinforcement tungsten carbide is added to Al2024 alloy through vortex method, the reinforcement is varied from 1% to 5% volume fraction of the matrix alloy, Highest values were recorded for hardness, tensile strength, compressive strength and young's modulus for addition of 3% volume fraction of reinforcement, beyond 3% these values start decreasing but the ductility of the composites decreases with addition of reinforcement for all the combinations. The increase in strength values because of the presence of hard WC particles which impart more strength to the base matrix and reduction in ductility was because of the presence of hard ceramic particles due to localized crack initiation and embrittlement to local stress concentration sites at the particle reinforcement interface [29]. Al2024 matrix reinforced with different weight fraction of gas atomised glass powder $\text{Fe}_{49.9}\text{Co}_{35.1}\text{Nb}_{7.7}\text{B}_{4.5}\text{Si}_{2.8}$ was fabricated using powder metallurgy. It is observed that tensile strength of the composite increased very significantly by 20% for unreinforced Al 2024 for glass powder reinforced Al2024 and yield strength of the composite increased by 21% for the pure alloy for the addition of 40% volume fraction of glassy particles due to the brittleness under tension of the matrix was reduced by incorporating a deformable crystalline phase glassy particles [25]. The microstructural evaluation of SiC particle reinforced Al 2024 composites interface was performed using Transmission Electron Microscopy (TEM) and strength conditions were evaluated by tensile modulus and hardness measurement and the results shows that interface condition of overall Al–SiC_p composites were good and three different types of interfaces were observed, which include immense majority of clean planar interfaces, minute amorphous interfaces and few slight reaction interfaces. These composites were also subjected to aging; aging was performed at 190 °C for 9 h after solution treating at 510 °C for the duration of 2 h. The hardness of the composite records highest value at this time because of innumerable discoid-shaped and needle-shaped nano sized precipitates [26]. In case of ceramic reinforcement

Table 4. Summary of materials, processing technique and its properties of Al 2xxx series alloys.

References	Canakci A <i>et al</i> [22]	Basavarajappa S <i>et al</i> [23]	Miyajima T <i>et al</i> [24]	Marko D <i>et al</i> [25]	Pei LI <i>et al</i> [26]	XIU ZY <i>et al</i> [27]
Matrix type	Al 2024	Al 2024	Al 2024	Al 2024	Al 2024	Al 2024
Reinforcement	B ₄ C particles	SiCp and graphite	SiC whiskers, Al ₂ O ₃ fibers and SiC particles	glass powder Fe _{49.9} Co _{35.1} Nb _{7.7} B _{4.5} Si _{2.8}	SiC particles	Si ₃ N ₄ particles
Reinforcement size	45–125 μ m	23 μ m of SiCp 45 μ m of Graphite	SiCw 0.3–1 μ m dia & 7 5–15 μ m in length Al ₂ O ₃ fibers 4 μ m dia & 40–200 μ m length 10 μ m diameter of SiCp	<30 μ m diameter	10 μ m diameter	5 μ m diameter
Volume fraction	Upto 12%	15% of SiC 3% of Graphite	5 to 29% of SiCw 3%–26% of Al ₂ O ₃ fibers 2%–10% of SiCp	10%–40% with increment of 10%	Upto 35%	45%, 50% and 55%
MMC Fabrication	Stir casting	Stir casting	Powder metallurgy	Stir casting	Powder metallurgy	Pressure infiltration method
Effect on its properties	density of the composite decreases by increasing reinforcing content, but hardness and porosity increases with increasing volume fraction of reinforcement particle	Al–SiC–Gr composites exhibit higher wear resistance compared with Al–SiC reinforcement composites. Wear loss decreases with an increase in sliding velocity for the hybrid composites, but increases with increase in load and sliding distance	whisker reinforcements have effectively reduced the steady state wear rates, fiber fragments of larger size caused aggressive action on the counter face material becomes heavier with increasing volume fraction, particle reinforcement, avoids initial severe wear and decreases the steady state wear rate considerably because of the large load supporting effect	tensile strength increased by 20% for glass powder addition and yield strength increased by 21% for the addition of 40 wt% of glassy particles due to the brittleness under tension of the matrix was reduced by incorporating a deformable crystalline phase glassy particles	hardness of the composite records highest value at this time because of innumerable discoid-shaped and needle-shaped nano sized precipitates	as the reinforcement content increases the bending strength of the composite increases, this is because interface of the composite becomes more stronger, where interface plays a major role in load transferring and it was also observed that heat treatment has lesser effect on the fracture surface of the composites

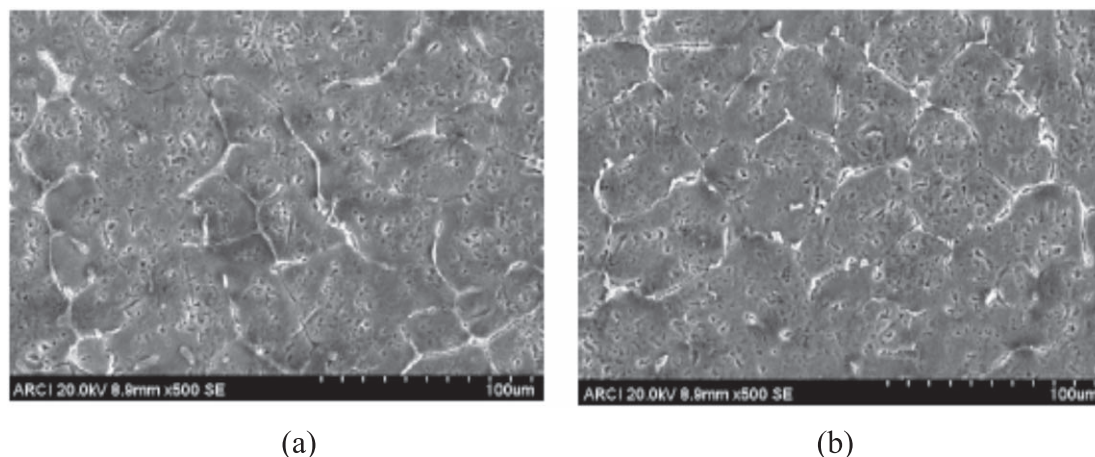


Figure 3. SEM images of (a) AA 2024–10% FA/SiC (b) AA 2024–15% FA/SiC [2].

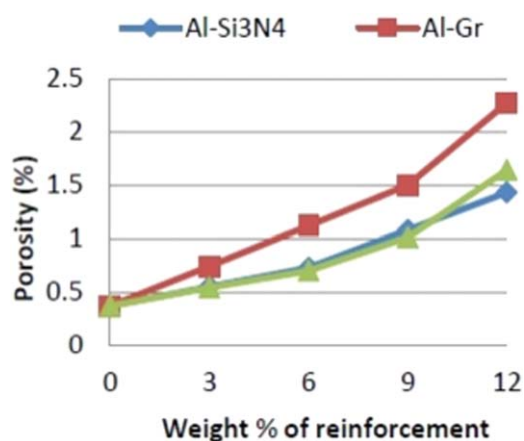


Figure 4. Variation of porosity with the weight % of the reinforcement [22].

Si₃N₄ to Al–Cu alloy matrix, the weight fraction of reinforcement is varied from 45% to 55% and fabricated using pressure infiltration method. In microstructural study the high density of dislocations in matrix alloys around the Si₃N₄ was found because of difference in thermal stress generation due to variation of co-efficient of thermal expansion between the reinforcement and matrix. Effect of reinforcement, T6 heat treatment on the bending strength of matrix was also studied and the results show that as the reinforcement content increases the bending strength of the composite increases, this is because interface of the composite becomes more stronger, where interface plays a major role in load transferring and it was also observed that heat treatment has lesser effect on the fracture surface of the composites [27]. The effect of particle size on the physical and mechanical properties of Al–Cu were carried out, the composites were fabricated by squeeze casting method and it was observed that as the particle size decreases the density of the composite decreases but porosity shows the reverse trend [30]. Table 4 summarises the materials, processing techniques and properties of Al 2xxx alloys.

2.3. Aluminium 3xxx series

In 3xxx series alloys manganese is the major constituent and generally non-heat treatable. The composition of commercially available aluminium 3xxx series is shown in the table 5.

The precipitation treatment at different temperatures for Al 3004 shows dispersion strengthening achieved by α -Al(MnFe)Si dispersoids. The yield strength in compression reaches its peak value due to a large volume fraction of fine and the dispersoids were found to be thermally stable at 300 °C upto 1000 h of holding, also leads to consistent high mechanical performance and creep resistance [31, 32]. Transmission Electron Microscopy study shows the size, type and distribution of α -Al(MnFe)Si dispersoids which strongly influence mechanical properties and deformation behaviour of Al 3003 alloys [33].

Table 5. Commercially available aluminium 3xxx series alloys [14].

Registered international designation	Composition in Wt%									
	Mn	Cu	Si	Cr	Mg	Ni	Zn	Ti	Pb	Zr
3003	1.2	0.12	—	—	—	—	—	—	—	—
3004	1.2	—	—	—	1	—	—	—	—	—
3005	1.2	—	—	—	0.40	—	—	—	—	—
3006	0.7	0.20	0.50	0.20	0.5	—	0.3	0.10	—	—
3008	1.5	0.10	0.40	0.05	0.01	0.05	0.05	0.10	—	0.40
3018	1.2	0.20	0.30	0.10	1.2	—	0.25	0.10	0.01	—
3105	0.60	—	—	—	0.50	—	—	—	—	—

Table 6. Commercially available aluminium 4xxx series alloys [14].

Registered international designation	Composition in Wt%							
	Si	Mn	Cu	Mg	Fe	Cr	Zn	Ti
4001	2.0	0.10	0.20	0.05	0.8	—	0.10	0.15
4002	4.0	0.03	0.10	0.10	0.35	—	0.15	0.02
4003	7.5	0.10	0.25	2.5	0.8	—	0.20	—
4005	10.5	0.10	0.25	0.6	0.8	—	0.20	—
4011	7.0	0.10	0.20	0.6	0.20	—	0.10	0.1
4012	12.0	0.7	1.5	0.75	0.8	—	1.0	—
4032	12.2	—	0.9	—	—	—	0.9	—
4043	5.2	—	—	—	—	—	—	—
4048	10.0	0.07	4.0	0.07	0.8	0.07	10.0	—
4643	4.1	—	—	0.20	—	—	—	—

2.4. Aluminium 4xxx series

The major alloying element in this series is silicon, Most of the alloys in this series are non-heat treatable, but when it is required to use in welding the heat treatable alloys this will be altered by removing some of the alloying constituents so it responds to heat treatment to a limited extent. The composition of commercially available aluminium 4xxx series is shown in the table 6.

The effect of δ -Al₂O₃ short fiber on Al–Si alloy was studied and its dry wear and friction properties were evaluated. Wear test results shows that increase in volume of fiber was found to decrease in wear rate and coefficient of friction because with the addition of alumina fiber the plastic deformation decreased as a result wear rate of the composite decreases. The coefficient of friction was found to be dependent on the applied load [34]. The comparative study was made between hybrid Aluminium composite Al–Si reinforced with activated carbon and mica with conventional composites Al–Si alloy reinforced with activated Carbon and Mica individually, it is found that hybrid composite exhibits increased hardness compared to single reinforced composites. This is due to the presence of hard particles which offer more resistance to plastic deformation which resulted in increase of hardness of the composite. Hybrid composite resulted in 29% and 14% increase in hardness than Al–Si-activated carbon and Al–Si-Mica respectively [35]. The dry sliding wear behaviour of Al–Si matrix reinforced with graphite and SiC and also the effect of sliding distance, load and sliding speed on the wear behaviour using ANOVA was observed, the reinforcement was added upto 10% of matrix weight. The experimental result shows that the wear rate reduces with increase in reinforcement percentage till 7.5% addition beyond this the wear rate starts decreasing. Thus it is concluded that optimal result can be found at 7.5% addition of reinforcement for any value of load, sliding distance and sliding speed. ANOVA results shows that sliding distance is the highest contributor for the wear loss followed by sliding speed and load [6]. The mechanical and machining properties of the composite Al–Si reinforced with SiC_p, the amount of reinforcement was varied from 5% to 15% of total volume fraction. It was observed that increase of reinforcement content produced better strength properties such as hardness and impact strength, but tensile strength showed a different trend; increased upto 10% addition of SiC_p and then decreased when reinforcement content increased to 15%, this decrement in tensile strength is due to broken particles in structure as observed from the microstructure study. Machinability of the developed composite are different from traditional materials because of the presence of the abrasive filler material, it causes more wear on cutting tools. Wear on flank portion also increased with increase in reinforcement content [36]. The Al–Si alloy was reinforced with SiC particles fabricated using spray forming process, the effect of reinforcement on its wear and frictional behaviour was evaluated. The microstructure study using SEM shows that spray forming process is effective in producing metal matrix composites and composites exhibits better wear resistance than the unreinforced alloys. The wear test conducted at four different temperatures 50 °C, 75 °C, 100 °C and 150 °C showed that with increase in temperature the wear rate of the

Table 7. Summary of materials, processing technique and its properties of Al 4xxx series alloys.

References	Akbulut H <i>et al</i> [34]	Sarada BN <i>et al</i> [35]	Ozben T <i>et al</i> [36]	Kaur K <i>et al</i> [37]	Radhika N <i>et al</i> [38]
Matrix type	Al–Si alloy	Al–Si ally	Al–Si ally	Al–Si alloy	Al–Si Alloy
Reinforcement	δ -Al ₂ O ₃ short fiber	Activated carbon and mica	SiC particles	SiC particles	boron carbide particles
Reinforcement size	—	—	30–60 μ m diameter	106–125 μ m diameter	33–40 μ m diameter
Volume fraction	10, 15, 20, 25, 30%	10%	5, 10 and 15%	5%	10%
MMC Fabrication	Modified liquid infiltration method	Stir casting	Pressure casting	Spray forming process	Stir casting process
Effect on its properties	increase in volume of fiber was found to decrease in wear rate and coefficient of friction because with the addition of alumina fiber the plastic deformation decreased as a result wear rate of the composite decreases	hybrid composite exhibits increased hardness compared to single reinforced composites, due to the presence of hard particles which offer more resistance to plastic deformation which resulted in increase of hardness of the composite	increase of reinforcement content produced better strength properties such as hardness and impact strength, but tensile strength increased upto 10% addition and then decreased for 15%, this decrement in tensile strength is due to broken particles in structure as observed from the microstructure study.	Composites exhibits better wear resistance than the unreinforced alloys. wear test conducted for the temperatures 50 °C, 75 °C, 100 °C and 150 °C, it showed that with increase in temperature the wear rate of the composites decreases	the analysis of variance results indicates that the load was the dominant parameter on abrasive wear behaviour

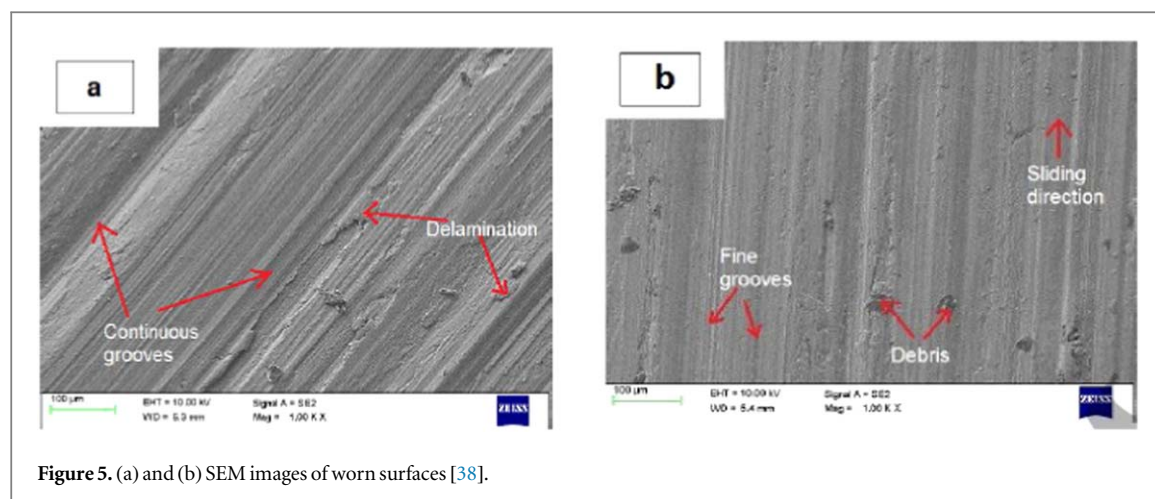


Figure 5. (a) and (b) SEM images of worn surfaces [38].

Table 8. Commercially available aluminium 5xxx series alloys [14].

Registered international designation	Composition in Wt%										
	Fe	Si	Mg	Mn	Cu	Cr	Ni	Zn	Ti	Pb	Zr
5002	0.15	0.10	1.2	—	0.15	—	—	—	—	—	—
5004	0.7	0.40	1.0	0.30	0.40	0.10	—	1.0	0.15	—	—
5007	0.006	0.006	0.5	0.001	0.006	0.001	0.001	0.003	0.002	—	—
5008	0.006	0.006	1.0	0.001	0.006	0.001	0.001	0.003	0.002	—	—
5009	0.006	0.006	2.0	0.001	0.006	0.001	0.001	0.003	0.002	—	—
5011	0.7	0.6	0.8	0.60	0.20	0.10	—	0.10	—	—	—
5012	0.10	0.08	1.8	0.05	0.10	—	—	0.03	0.03	—	—
5013	0.25	0.20	3.5	0.40	0.03	0.03	0.03	0.10	0.10	0.03	0.05
5014	0.40	0.40	4.8	0.50	0.20	0.20	—	1.2	0.20	—	—
5015	—	—	0.8	0.08	0.08	—	—	—	0.02	—	—
5020	0.7	0.30	2.8	0.30	1.5	0.20	—	0.20	0.10	—	—
5025	0.25	0.25	5.2	0.20	0.10	0.20	—	0.25	0.10	—	0.20
5034	0.60	0.40	0.80	0.35	0.40	0.10	—	0.40	0.10	—	—
5039	0.40	0.10	3.8	0.40	0.03	0.15	—	2.8	0.10	—	—
5053	—	—	3.5	0.07	0.07	—	—	—	0.02	—	—
5055	0.7	0.30	4.5	0.50	0.25	0.25	—	0.25	—	—	—
5080	—	—	4.0	0.50	0.10	0.10	—	2.5	0.20	—	—
5085	0.40	0.30	6.2	0.20	0.15	0.20	—	0.20	0.10	—	—
5090	0.50	0.50	7.0	0.35	0.25	0.15	—	0.20	0.02	—	—

composites decreases [37]. The incorporation of coated steel fibers to Al–Si alloys processed using stir casting process shows improved hardness, the analysis of variance results shows that reinforcement percentage and stirring speed are the dominant parameter on abrasive wear behaviour. Figure 5(a) and (b) shows the microstructure observations of worn surfaces, long continuous grooves on the surface parallel to its sliding direction and delamination type of wear were observed [38]. Hardness and dry sliding behaviour of heat treated Fe-rich Al–Si alloys is improved with the addition of bifilm oxides, it was observed that the bifilm oxides expands during heat treatment and exhibits higher wear strength than the non-heat treated alloys [39]. Table 7 summarises the materials, processing techniques and properties of Al 4xxx alloys.

2.5. Aluminium 5xxx series

Magnesium is the major alloying element in 5xxx series, the composition of commercially available aluminium 5xxx series is shown in the table 8.

In order to improve the surface properties by retaining its structural properties the Al5083 alloys were reinforced with alumina and TiB₂ processed using *in situ* process. The incorporation of reinforcement on the matrix alloy were observed using XRD pattern shown in figure 6 [40]. The wear rate of the composite, approximately 40% lower than the base alloy was observed by incorporating these boron carbide particles about 5% of the total volume fraction and it proves that this particle has a significant influence on the wear behaviour of the base alloy [41]. Improved wear behaviour was recorded with addition of SiC particles in Al–Mg alloy, the

Table 9. Summary of materials, processing technique and its properties of Al 5xxx series alloys.

References	Yuvaraj N <i>et al</i> [40]	Tang F [41]	Gargatte S [42]	Idri AH [43]	Rana S [44]	Hariprasad T [45]
Matrix type	Al 5083	Al 5083	Al 5083	Al 5083	Al 5083	Al 5083
Reinforcement	B ₄ C particles	B ₄ C particles	SiC particles	SiC particles	SiC particles	B ₄ C & Al ₂ O ₃
Reinforcement size	30–60 μ m diameter	4 μ m diameter	1–25 μ m diameter	29.2 μ m diameter	35 μ m diameter	Mesh size 400 μ
Volume fraction	—	5 and 10%	3, 5 & 7%	0, 3, 5, 8, 10%	3, 5, 8 and 10%	0, 8, 10 and 13%
MMC Fabrication	Friction stir processing	Cryomilling and consolidating process	Stir casting process	Ultrasonic assisted stir casting process	Ultrasonic assisted stir casting process	Vortex method
Effect on its properties	the surface composite layer lead to three passes with nano reinforcement particle and exhibited enhanced properties in hardness, the variation of micro hardness for different number of passes	wear rate of the composite decreases by 40% than the base alloy was observed by incorporating boron carbide particles about 5%	wear resistance increased considerably with addition of particles, because the addition of SiC increases the hardness of the composite there by it restricts the deformation of matrix material with increasing load	density and porosity of the composite increase with the addition of reinforcement, and hardness of the composite increases with reinforcement content	application of ultrasonic vibration on the composite at the time of melting not only refined the grain structure of the composite it also improves the distribution of nano-sized particulates	combined effect of Al ₂ O ₃ and B ₄ C on Al–Mg based composites shows that the hybrid composites have better wear resistance,

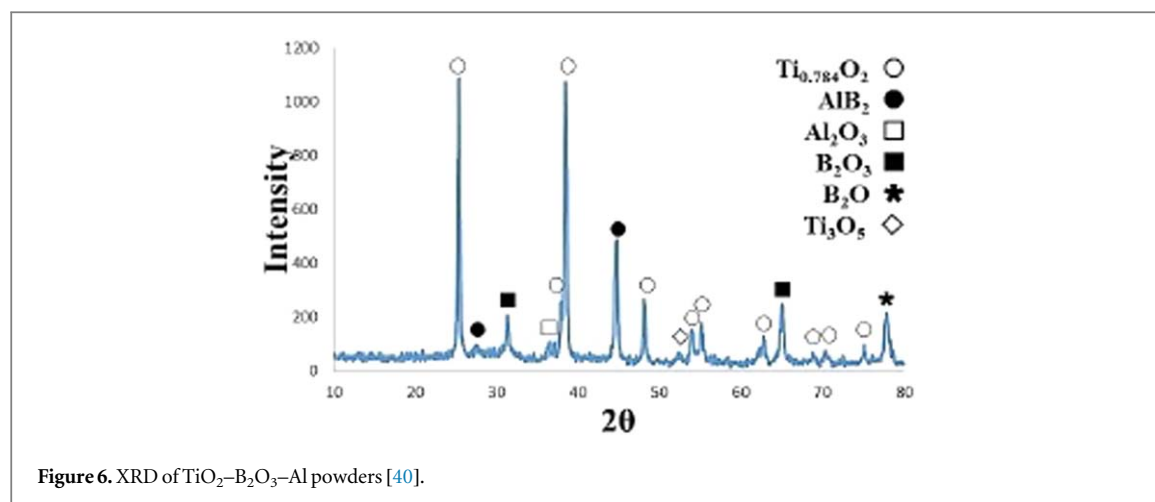


Figure 6. XRD of $\text{TiO}_2\text{-B}_2\text{O}_3\text{-Al}$ powders [40].

Table 10. Commercially available aluminium 6xxx series alloys [14].

Registered international designation	Composition in Wt%										
	Si	Mn	Cu	Mg	Fe	Cr	Zn	Ti	Pb	Sn	Zr
6001	0.50	0.03	0.10	0.50	0.6	0.03	0.10	—	—	—	—
6004	0.40	0.40	0.10	0.50	0.20	—	0.05	—	—	—	—
6007	1.20	0.15	0.20	0.8	0.7	0.10	0.25	0.15	—	—	0.10
6017	0.60	0.10	0.10	0.60	0.20	0.10	0.05	0.05	—	—	—
6030	0.60	0.15	0.25	1.0	0.70	0.20	0.25	0.15	—	0.30	—
6051	0.80	0.20	0.15	0.60	1.0	—	0.25	—	—	—	—
6061	0.60	0.15	0.25	1.0	0.7	0.20	0.25	0.15	0.003	—	—
6063	0.40	0.10	0.10	0.8	0.35	0.10	0.10	0.10	—	—	—
6070	1.40	0.8	0.25	0.8	0.50	0.10	0.25	0.15	—	—	—
6091	0.60	0.15	0.25	1.0	0.70	0.15	0.25	0.15	—	—	—
6111	0.85	0.28	0.70	0.75	—	—	—	—	—	—	—

volume fraction of SiC particles are varied from 3% to 7%. It was observed that wear properties are considerably increased with addition of particles, because the addition of SiC increases the hardness of the composite there by it restricts the deformation of matrix material with increasing load. Statistical results shows that sliding speed has major influence on wear rate and sliding distance has lesser influence on wear rate [42]. The effect of ultrasonic probe assistance on fabricating SiC reinforced Al–Mg composites using stir casting method was made and also its effects on physical and mechanical properties were observed. The result shows that the density and porosity of the composite increase with the addition of reinforcement, it was also observed that hardness of the composite increases with reinforcement content and it was concluded that ultrasonic stir casting route to be a prominent route for fabrication of particulate reinforced metal matrix composite [43]. Elastic modulus of the composite increased by about 8.6% with the addition of SiC particle upto 10% of the total weight fraction. The micrographs obtained from SEM shows uniform distribution of SiC particles. The application of ultrasonic vibration on the composite at the time of melting not only refined the grain structure of the composite it also improves the distribution of nano-sized particulates [44]. The combined effect of Al_2O_3 and B_4C on Al–Mg based composites shows that the hybrid composites have better wear resistance, the wear rate decreases for hybrid composite compared to the single reinforced composites [45]. Table 9 summarises the materials, processing techniques and properties of Al 5xxx alloys.

2.6. Aluminium 6xxx series

This series contain silicon and magnesium nearly in the proportions that are required for the formation of magnesium silicide (Mg_2Si), thus resulting a heat treatable alloy. The composition of commercially available aluminium 6xxx series is shown in the table 10.

The alloy Al6061 is most commonly used among other alloys in 6xxx series, the strength properties of this alloy can be further enhanced by reinforcing it with ceramic reinforcement. The effect of SiC particle content and its grain size on this alloy was observed for its tribological behaviour, the SiC particles content varied from 5% to 40% of the total volume fraction and SiC particle of 150 to 600 of mesh size was used and it was concluded that wear rate can be minimized with addition of reinforcement, the better results were obtained with the addition of 35% of its total volume fraction and 600 mesh size of SiC [46]. The effect of SiC nano particles on the same matrix shows more efficient

Table 11. Summary of materials, processing technique and its properties of Al 6xxx series alloys.

References	Mishra AK <i>et al</i> [46]	Knowles <i>et al</i> [12]	Christy <i>et al</i> [47]	Ramesh CS <i>et al</i> [48]	Pramanik A <i>et al</i> [49]	Ramesh A <i>et al</i> [50]	Narimani M <i>et al</i> [51]
Matrix type	Al 6061	Al 6061	Al 6061	Al 6061	Al 6061	Al 6061	Al 6063
Reinforcement	SiC particles	SiC nano particles	TiB ₂	TiB ₂	Al ₂ O ₃	Albite and graphite	B ₄ C and TiB ₂
Reinforcement size	Mesh size 150 to 600 microns	500 nm	—	—	6–18 μ m diameter	Mesh size of 50 microns	7 μ m of B ₄ C 40–60 μ m of TiB ₂
Volume fraction	5 to 35% with increment of 5%	15%	12%	5, 7.5 & 10%	10%	0, 1, 2 and 4%	10%
MMC Fabrication	Stir casting	Powder metallurgy	<i>in situ</i> process	<i>In-situ</i> process	Stir casting	Vortex method	Powder metallurgy
Effect on its properties	wear rate can be minimized with addition of reinforcement, the better results were obtained with the addition of 35% of its total volume fraction and 600 mesh size of SiC	nano SiC particles shows more efficient strengthening than the volume fraction of larger grain size particles, due to more number of barriers to dislocations and the particles do not fracture at the time of deformation	improves the hardness, young's modulus and tensile strength with loss of ductility	effect of <i>in situ</i> TiB ₂ shows significant improvements in the hardness, a minimum of 9.077% to 22.26%. The improvement in ultimate tensile strength was also observed because of uniform distribution of reinforcement particles and reaction between the alloy and reinforcement	enhance the wear properties by resisting deformation and stopping plastic deformation, avoiding large wear fragments from becoming separated by gross seizure and abrasive wear. This is because of the formation of oxide film on the metal surface providing a support, resulting in hardening and strengthening of the interfacing material	Al6061-Albite composites shows a significant increase in its hardness, tensile strength and young's modulus but reduction in its ductility property, where Al6061-Graphite composites shows a improvement in its ductility, tensile strength, compressive strength and elastic modulus but a decrease in its hardness	embedment of these particles on the surface increased the hardness and wear properties

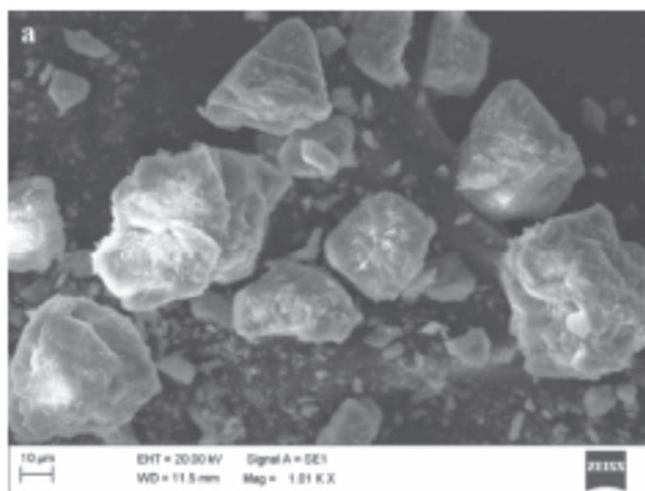


Figure 7. SEM images of Al_2O_3 [48].

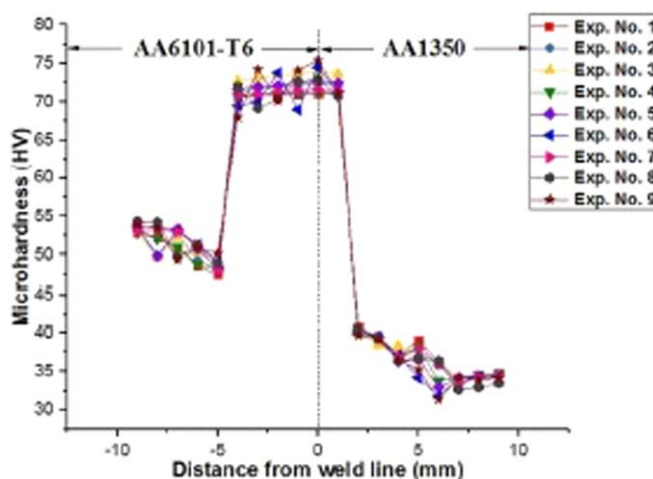


Figure 8. Variation of Hardness with respect to distance from wild line [48].

strengthening of the composites than a same volume fraction of larger grain size particles, due to more number of barriers to dislocations and the particles do not fracture at the time of deformation [12]. The inclusion of TiB_2 as the reinforcement in the Al6061 using stir casting process, it was observed that addition of reinforcement improves the hardness, young's modulus and tensile strength with loss of ductility [47]. The effect of Al_2O_3 reinforcement on dissimilar aluminium matrix shows significant improvements in the hardness with addition of reinforcement, SEM image of Al_2O_3 reinforcement as shown in figures 7 and 8 enables efficient transfer of load from matrix to reinforcement. It results in longer duration to initiate a crack at the interface [48]. The alumina reinforced metal matrix composites enhance the wear properties by resisting deformation and stopping plastic deformation that happens in the vicinity of the particle, avoiding large wear fragments from becoming separated by gross seizure and abrasive wear. This happens because of the formation of oxide film on the metal surface providing a support, this results in hardening and strengthening of the interfacing material [49]. The same matrix was reinforced with boron carbide fabricated using stir casting technique, it shows improved hardness with the addition of reinforcement [52]. The influence of *in situ* formed ZrB_2 particles on the Al6061 was observed, microstructural observation shows uniform distribution of particles in the matrix and it directly relates the solidification process of the matrix alloy. The microhardness and macrohardness of the composite increases with the addition of reinforcement [11]. The comparative study on the effect of albite which is naturally available plagioclase feldspar and graphite reinforced aluminium alloy produced using vertex method of manufacturing, different trend was observed with use of these reinforcements. Al6061-Albite composites shows a significant increase in its hardness, tensile strength and young's modulus but reduction in its ductility property, where Al6061-Graphite composites shows an improvement in its ductility, tensile strength, compressive strength and elastic

Table 12. Commercially available aluminium 7xxx series alloys [14].

Registered international designation	Composition in Wt%										
	Si	Fe	Mn	Cu	Mg	Ti	Zn	Cr	Pb	V	Zr
7003	0.30	0.35	0.30	0.20	0.80	0.20	5.50	0.20	—	—	0.10
7004	0.25	0.35	0.45	0.05	1.50	0.05	4.20	0.05	—	—	0.15
7010	0.12	0.15	0.10	1.80	2.40	0.06	6.20	0.05	—	—	0.15
7015	0.20	0.30	0.10	0.10	1.80	0.10	5.80	0.15	—	—	0.15
7019	0.35	0.45	0.30	0.20	2.0	0.15	4.50	0.10	—	—	—
7029	0.10	0.12	0.03	0.75	1.70	0.05	4.50	—	—	0.05	—
7035	0.15	0.25	0.10	0.20	3.0	0.03	5.0	0.05	—	—	0.15
7050	0.12	0.15	0.10	2.20	2.20	0.06	6.20	0.04	—	—	0.12
7065	0.06	0.08	0.04	2.10	1.60	0.06	7.60	0.04	—	—	0.10
7075	0.40	0.50	0.30	1.50	2.50	0.20	5.50	0.25	—	—	—
7079	0.30	0.40	0.20	0.60	3.20	0.10	4.0	0.15	—	—	—
7090	0.12	0.15	—	1.0	2.50	—	8.0	—	—	—	—
7099	0.10	0.12	0.04	1.80	1.80	0.06	8.0	0.04	—	—	0.10

modulus but a decrease in its hardness [50]. Al6061-SiC-Al₂O₃ hybrid composites fabricated using stir casting process are tested for its tribological behaviour, in addition to reinforcement content the influence of applied load, rotational speed are also observed, the analysis shows among all the factors volume fraction has highest significance which affects the wear behaviour [53]. Hybrid surface composite layers of Al 6063 matrix containing boron carbide and titanium boride particles using friction stir processing, the results show that embedment of these particles on the surface increased the hardness and wear properties compared to the base alloy [51]. Table 11 summarises the materials, processing techniques and properties of Al 6xxx alloys.

2.7. Aluminium 7xxx series

In this series Zinc is the major alloying element, it is added in amounts of 1 to 8% and it is coupled with a smaller percentage of magnesium results in a heat treatable alloy. The composition of commercially available aluminium 7xxx series is shown in the table 12.

Al7075 aluminium alloy matrix reinforced with boron nitride and zirconium oxide combination shows good wettability between matrix and reinforcement as it is confirmed in SEM analysis as shown in figure 9 and it found applications in the field of aircraft and space industries due to its lower weight to strength ratio, good wear resistance [54]. The same composite fabricated using squeeze casting technique shows an improved hardness and flexural strength compared to unreinforced matrix alloys. It was found that 10% addition of SiC and particle size of 29 μm resulted in 40% increase in its flexural strength and 30% enhancement in its hardness value [55]. The effect of stirring speed on microstructure of the composites were observed, experiments were carried out for 500, 650 and 750 rpm of stirring speed for 10 min and the results reveal that the composites produced at 650 rpm had uniform distribution of SiC particles [60]. The influence of Al₂O₃ on the Al7075 matrix alloy shows an enhanced impact strength that leads to use in shock loading environment, both in as cast and forged applications [61]. Grain size of Al₂O₃ has the major influence on the hardness properties of the composite compared to other parameters, the addition of 15 wt% Al₂O₃ and grain size of 36 μm exhibited highest microhardness [56]. The influence of boron carbide on the mechanical and tribological behaviour of Al7075 shows increased hardness compared to the base alloy, this increase corresponds to the presence of hard ceramic phase. The wear resistance of the composite improved with the addition of reinforcing content, a mechanically mixed layer containing iron and oxygen acts as an effective layer in preventing metal to metal contact [57]. Al7075 reinforced with graphite shows an enhanced structural and wear properties, the graphite content increased from 5 to 10 wt% of its total volume fraction. It is observed that the addition of 5 wt% boron carbide resulted in less wear rate and same trend was observed for hardness and flexural strength [62]. The most inexpensive and low density material flyash can be used to overcome the cost barrier, the incorporation of this to the aluminium alloy shows an improved toughness, hardness and tensile strength. Density of the composites also decreased considerably and can be used in the applications where weight plays an important role [63]. The short basalt fibre reinforced with aluminium alloys fabricated using compo casting method, the results shows that as the reinforcement content increases the hardness, ultimate tensile strength and young's modulus of the composite increases with a reduction in ductility [58]. The effect of graphite and alumina particles on the hardness of the hybrid composites shows an improvement with increasing ceramic reinforcement content. Hardness, flexural strength and compression strength of the hybrid composite decreases with the addition of graphite and it was overcome by the addition of alumina particle [59]. The use of waste sugarcane bagasse ash as a reinforcement with graphite shows significant

Table 13. Summary of materials, processing technique and its properties of Al 7xxx series alloys.

References	Bhushan RK <i>et al</i> [54]	Kalkanli <i>et al</i> [55]	Deshamanya IB <i>et al</i> [56]	Baradeshwaran A <i>et al</i> [57]	Karthigeyan R <i>et al</i> [58]	B Swathi <i>et al</i> [59]
Matrix type	Al 7075	Al 7075	Al 7075	Al 7075	Al 7075	Al 7075
Reinforcement	SiC particles	SiC particles	Al ₂ O ₃ particles	B ₄ C particles	Short Basalt Fiber	Al ₂ O ₃ particles and graphite
Reinforcement size	20– μ m diameter	29 μ m diameter	36, 45, 54, 63, 72 μ m diameter	16–20 μ m diameter	1.5 to 2 mm	0%–3% Al ₂ O ₃ 2%–6% graphite
Volume fraction	10% and 15%	10, 15, 20 and 30%	—	5, 10, 15 and 20%	0%–6%	—
MMC Fabrication	Liquidus semi continuous casting	squeeze casting technique	Stir casting	Stir casting	Stir casting	Stir casting
Effect on its properties	addition of 10% to 15 wt% SiC particles and grain size of 20–40 μ m can be used for applications where temperature does not exceed 1250 °C.	10% addition of SiC and particle size of 29 μ m resulted in 40% increase in its flexural strength and 30% enhancement in its hardness value	Grain size of Al ₂ O ₃ has the major influence on the hardness properties of the composite compared to other parameters, the addition of 15 wt% Al ₂ O ₃ and grain size of 36 μ m exhibited highest microhardness	wear resistance of the composite improved with the addition of reinforcing content, a mechanically mixed layer containing iron and oxygen acts as an effective layer in preventing metal to metal contact	as the reinforcement content increases the hardness, ultimate tensile strength and young's modulus of the composite increases with a reduction in ductility	Hardness, flexural strength and compression strength of the hybrid composite decreases with the addition of graphite and it was overcome by the addition of alumina particle

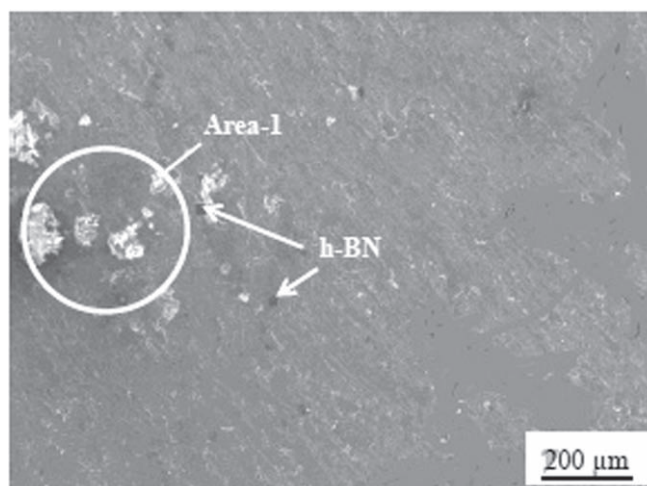


Figure 9. SEM micrographs of Al7075 + 3% h-BN + 4% ZrO₂ [54].

Table 14. Commercially available aluminium 8xxx series alloys [14].

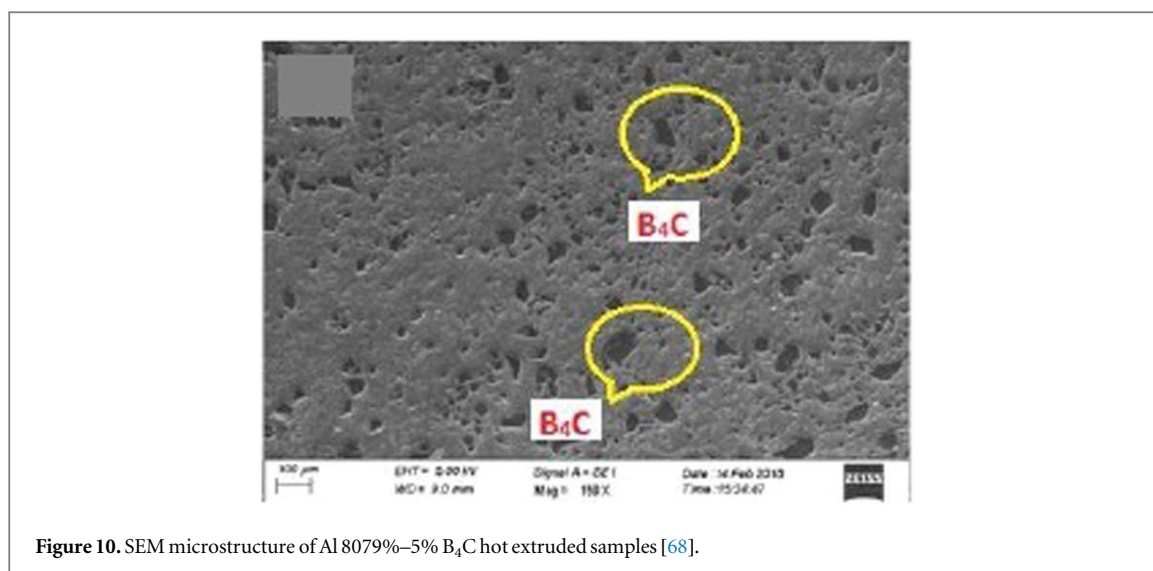
Registered international designation	Composition in Wt%										
	Si	Cu	Fe	Mg	Mn	Zn	Cr	Ti	B	Li	Zr
8005	0.40	0.05	0.60	0.05	—	0.05	—	—	—	—	—
8006	0.40	0.30	1.50	0.10	0.60	0.10	—	—	—	—	—
8007	0.40	0.10	1.50	0.10	0.60	1.4	—	—	—	—	—
8010	0.40	0.20	0.50	0.30	0.60	0.40	0.20	0.10	—	—	—
8011	0.70	0.10	0.65	0.05	0.20	0.10	0.05	0.08	—	—	—
8017	0.10	0.15	0.60	0.03	—	0.05	—	—	0.04	0.03	—
8019	0.20	—	8.50	—	0.05	0.05	—	0.05	—	—	—
8021	0.15	0.05	1.50	—	—	—	—	—	—	—	—
8025	0.10	0.20	0.10	0.05	0.05	—	0.18	0.001	—	—	0.10
8030	0.10	0.25	0.50	0.05	—	0.05	—	—	0.01	—	—
8050	0.20	0.05	1.15	0.05	0.50	0.10	0.05	—	—	—	—
8076	0.10	0.04	0.80	0.10	—	0.05	—	—	—	—	—
8079	0.15	0.05	1.0	—	—	0.10	—	—	—	—	—
8090	0.20	1.20	0.30	1.0	0.10	0.25	0.10	0.10	—	2.50	0.10

criteria for the manufacturing of the composite. Effect of this reinforcement on the Al7075 based hybrid metal matrix composites shows an improved hardness, yield strength and ultimate tensile strength [64]. Table 13 summarises the materials, processing techniques and properties of Al 7xxx alloys.

2.8. Aluminium 8xxx series

This series of alloys consisting of miscellaneous compositions, these may contain considerable amount of tin, iron, lithium and nickel, each constituents used for the particular characteristics. The composition of commercially available aluminium 8xxx series is shown in the table 14.

The effect of different particle size of SiC on Al8011 alloy fabricated through stir casting process are tested for its strength properties, the results proved that the highest hardness, tensile strength, yield strength can be obtained by addition of 63 μm size SiC particle and 6% of its total volume fraction. This increase in strength properties were due to increase in density of dislocations caused by the addition of SiC particle reinforcement. Ductility of the composite decreases with the addition of reinforcement particle and it recorded lowest value for 8% addition compared to 2% addition of reinforcement. Toughness of the composites increases with decrease in particle size even for different volume fraction [65]. Strength to weight ratio of the Al8011 alloys can be reduced by reinforcing micro boron carbide particles and red mud, hardness and tensile strength of the composite is improved with the addition of redmud particles and microstructure study confirms the uniform distribution of reinforced particles [66, 67]. The influence of boron carbide on Al8079 shows an increased wear rate than the unreinforced alloy because of higher hardness and wear rate value of the reinforcement, but the SEM study



shows the fair distribution of the boron carbide within the matrix as shown in the figure 10 [68]. Table 15 summarises the materials, processing techniques and properties of Al 8xxx alloys.

3. Applications of aluminium matrix composites

Applications of Aluminium matrix composites reinforced with ceramic reinforcements mainly found in the field of automobile and aircraft structures due its mechanical properties such as higher specific strength, better wear resistance and specific modulus [69–71], addition to this AMMCs are using in electronic packaging because of its excellent thermo-physical properties [72, 73]. The SiCp reinforced aluminium matrix composites are used in the development of precision instruments such as advanced space optics, including ultra-lightweight telescope because of its lower density [74]. The SiCp reinforced with Al 2014 and Al 7075 alloy matrix shows an excellent wear resistant applications [75]. Table 16 summarises the typical material properties and applications of aluminium alloys series incorporated with different types of reinforcement.

4. Conclusion

From the literature it is evident that aluminium metal matrix composites have found large applications in aerospace, automobile and other industrial applications because of their high strength to weight ratio.

The present review focused on the current trends in the field of Aluminium based metal matrix composites in terms of their processing techniques, mechanical, tribological and metallurgical properties. In this article, the influence of different types of reinforcements on aluminium alloy of all the series has been studied in detail. The different processing techniques developed to fabricate the metal matrix composite like stir casting, powder metallurgy, pressure infiltration process etc were also reviewed. The studies show that matrix composition, reinforcement type and their distribution and interfacial bonding between the matrix and reinforcement plays a crucial role in deciding the performance of the composite and it is found that stir casting technique is most economical and robust process to manufacture aluminium metal matrix composite in large scale where other techniques are limited to only a laboratory scale. The aluminium series of 2xxx, 5xxx, 6xxx and 7xxx are the most commonly used alloys because of its good corrosion resistance, high toughness and good weldability, with the addition of SiC, Alumina, TiB₂, ZrB₂, B₄C reinforcements will improve its surface properties and wear resistance of the alloys.

The aluminium alloy series of 7xxx is found to be the strongest alloy among all the series, in this 7075 and 7079 found large applications currently in aircraft structure and space industries because of its excellent mechanical properties, but these alloys have some drawbacks such as lesser wear resistance, temperature resistance and corrosion resistance. These properties can be enhanced by the addition of ceramic reinforcements like boron carbide, aluminium nitride, tungsten carbide etc. The data available for addition of these reinforcements needs further investigations, so there is future scope for evaluation of the influence of these reinforcements on the 7xxx series alloys.

Table 15. Summary of materials, processing technique and its properties of Al 8xxx series alloys.

References	Nagaraj Ashok <i>et al</i> [65]	Karthikkumar C <i>et al</i> [66]	Reenivasrao K. V <i>et al</i> [67]	Magibalan <i>et al</i> [68]
Matrix type	Al 8011	Al 8011	Al 8011	Al 8011
Reinforcement	SiC	B ₄ C and Redmud	Redmud	Fly ash
Reinforcement size	63, 76 & 89 μm diameter	B ₄ C— < 10 μm Redmud of 150 μm	90 μm	—
Volume fraction	2 and 4%	1%–5%	2%–20% with increment of 2%	12%
MMC Fabrication	Stir casting	Stir casting	Stir casting	Stir casting
Effect on its properties	highest hardness, tensile strength, yield strength recorded with addition of 63 μm size SiC particle and 6% volume fraction. Because of increase in density of dislocations caused by the addition of SiCp. Ductility of the composite decreases with the addition of SiC and it recorded lowest value for 8% addition.	Strength to weight ratio of the Al8011 alloys can be reduced by reinforcing micro boron carbide particles and red mud, hardness and tensile strength of the composite is improved with the addition of redmud particles	hardness and tensile strength of the composite is improved with the addition of redmud particles	shows an increased wear rate than the unreinforced alloy because of higher hardness and wear rate value of the flyash

Table 16. Applications of aluminium matrix composites.

Materials	Properties	Application
Al 1xxx series alloys reinforced with Al_2O_3 , Al_3Zr micro particles & Graphite	Extremely high corrosion resistance, formability and electrical conductivity.	Foil and strip for packaging, chemical processing equipment, nameplates, flue lining reflectors, fan blades, heat exchangers, electrical conductors, capacitors, spin hollowware and fin stock [16, 19, 76].
Al 2xxx series alloys reinforced with SiCp, WC, B_4C , Si_3N_4 , glassy powder and SiC whisker	High strength especially at elevated temperatures, toughness, weldability.	Precision instruments, Truck and aircraft wheels, aircraft fuel usage, truck suspension parts and wing skins, the parts requiring more strength at high temperature up to 150 °C that are more commonly used for fasteners and screw machine stock [74, 75]
Al 3xxx series alloys	Strain hardenable, high formability, offer more resistance to corrosion and are readily welded, brazed and soldered.	Beverage cans, furniture, storage tanks, highway signs, cooking utensils, chemical equipments, heat exchangers for commercial power plant and other architectural applications [31–33, 76]
Al 4xxx series alloys reinforced with SiCp, B_4C , $\delta\text{-Al}_2\text{O}_3$ fiber, Bifilm oxides and aluminium borate whiskers	Excellent flow characteristics, low co-efficient of thermal expansion and high resistance to wear	Architectural applications, forged engine piston, automotive and structural applications [71].
Al 5xxx series alloys reinforced with Al_2O_3 , SiCp, B_4C	Moderately high strength, excellent corrosion resistance, high toughness even at cryogenic temperature to near absolute zero	Marine applications, building and construction, highway structures including bridges storage tanks [40–43, 77].
Al 6xxx series alloys reinforced with SiCp, B_4C , TiB_2 MMCs	Good formability, machinability, weldability, high strength coupled with excellent corrosion resistance, high thermal conductivity.	Electronic packaging, Precision instruments, Automotive applications, truck, marine frames, railroads and pipelines etc [72, 74].
Al 7xxx series alloys reinforced with SiCp, Graphite, flyash and short basalt fiber	Better combination of strength, fracture toughness and resistance to corrosion	Large structural members in aircraft, automotive components, drill pipe etc [78].
Al 8xxx series alloys reinforced with SiCp, B_4C	High strength with smaller loss in electrical conductivity, Lower density higher stiffness	Aircraft components such as upper or lower wing, fuselage, empennage [66, 67, 76]

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ORCID iDs

V Mahesh Kumar  <https://orcid.org/0000-0002-2541-3560>

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